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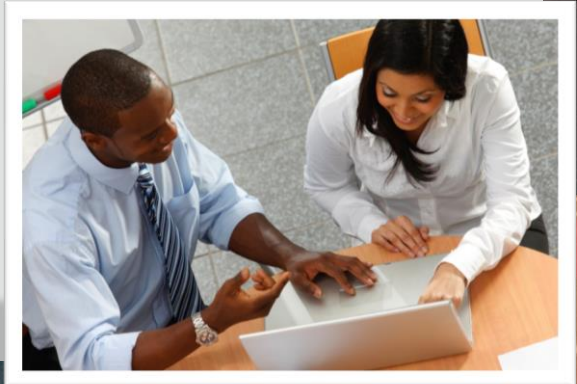
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Database Foundations

4-2

Convert a Logical Model to a Relational Model

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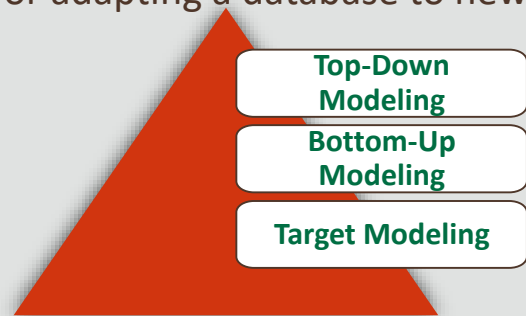
Objectives

- This lesson covers the following objectives:
 - Describe how to convert a logical model to a relational model in Oracle SQL Developer Data Modeler
 - List the steps to convert a logical model to a relational model
 - List the steps to convert a relational model to a logical model in Oracle SQL Developer Data Modeler



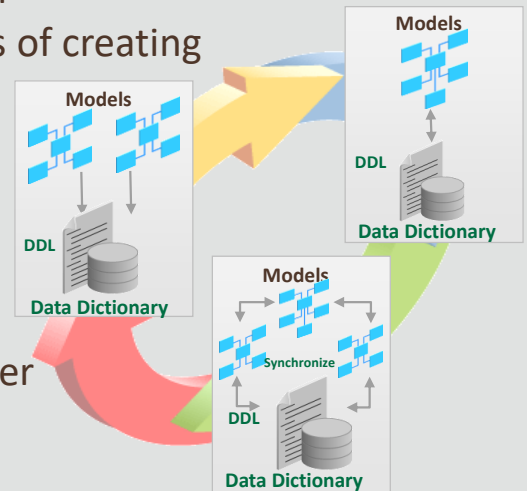
Data Modeling Approaches

- There are three approaches to data modeling:
 - Top-down modeling is the approach taken for designing a new database
 - Bottom-up modeling is the approach taken for creating a database based on extracting metadata from an existing database or using the data definition language (DDL) code obtained from an implementation of an existing database
 - Target modeling is best suited for adapting a database to new requirements



Engineering a Data Model

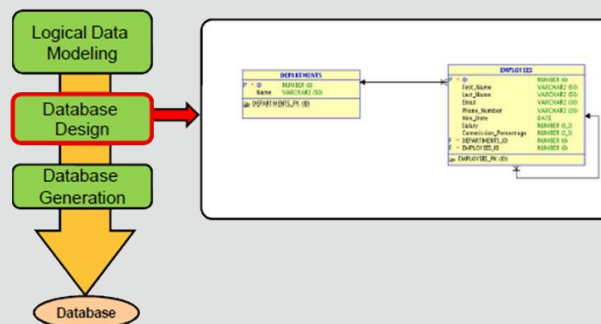
- Forward engineering is the process of transforming a logical data model to a relational model.
 - In Oracle SQL Developer Data Modeler a physical model is represented by a Relational Model
 - Reverse engineering is the process of creating a conceptual or logical model by extracting the information from an existing data source
 - In the next slides, you will see how to engineer a logical model to a relational model by using Oracle SQL Developer Data Modeler



Benefits of Creating a Relational Model

- A relational model:
 - Is closer to the implementation solution
 - Facilitates discussion
 - Forms the basis for the physical database design
- The ideal model can be adapted to a relational database management system (RDBMS) model

Information Requirements Relational Model



Overview of Relational Database

Table: MEMBERS

ID	FIRST_NAME	LAST_NAME	STREET_ADDRESS	CITY	STATE	ZIP
101	Jones	Smith	12 Oxford Street	Hudson	MA	01234

Primary Key

Columns

Table in the relational model

Row

MEMBERS		
P	* Id	NUMBER
	* First_Name	VARCHAR2 (30)
	* Last_Name	VARCHAR2 (30)
	* Street_Address	VARCHAR2 (50)
	* City	VARCHAR2 (30)
	* State	VARCHAR2
	* ZIP	VARCHAR2 (30)
MEMBERS_PK (Id)		

Terminology Mapping

ANALYSIS	DESIGN
Logical Model	Relational Model
Entity	Table
Attribute	Column
Instance	Row
Primary UID	Primary Key
Secondary UID	Unique Constraint
Relationship	Foreign Key
Business Constraints	Check Constraints

Some of the business rules are translated into check constraints. Other complex rules require additional programming. This initial mapping is limited to the design of tables, columns, and constraints that can be declared. A declarative constraint is a business constraint that can be ensured at the server level by using only database language statements; a declarative constraint requires no coding.

Case Scenario: Building a Relational Model



Faculty

Sean, is it possible to create a relational model from an existing logical model by using the Oracle SQL Developer Data Modeler tool?

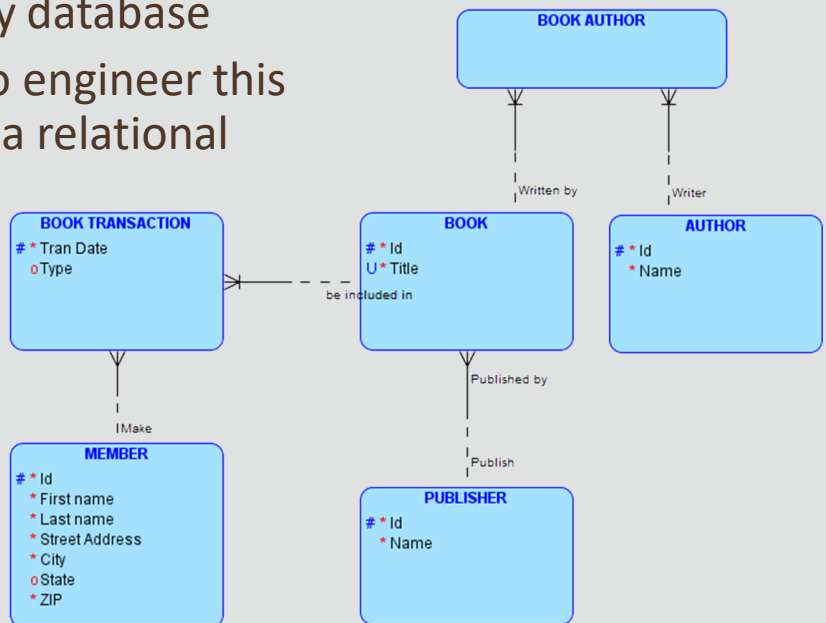
Absolutely. Let me list the steps involved in forward engineering a logical model to a relational model by using the Oracle SQL Developer Data Modeler tool.



Student

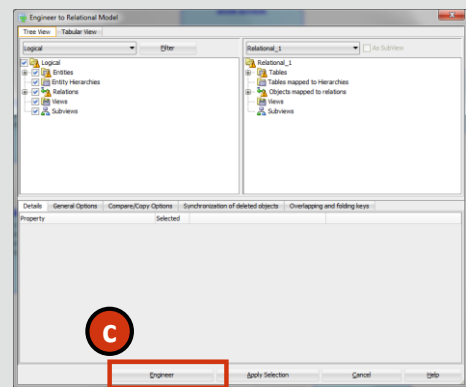
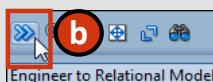
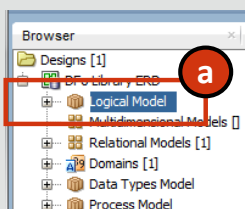
Case Scenario: Simplified Library Database ERD

- Sean has put together the logical model (ERD) for a simplified Library database
- Now he needs to engineer this logical model to a relational model by using the Oracle SQL Developer Data Modeler tool



Engineering a Relational Model

- Here are the steps involved in forward engineering a Logical Model to a Relational Model using Oracle SQL Developer Data Modeler:
 - a. Select the logical model.
 - b. Click the Engineer to Relational Model icon
 - c. Accept all the defaults and click Engineer



(Continued on next slide)

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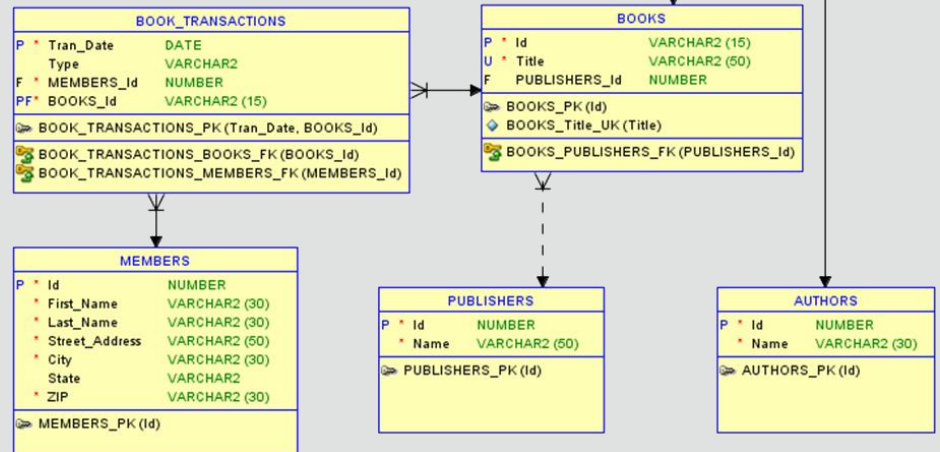
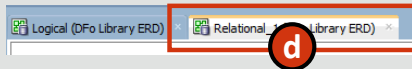
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Engineering a Relational Model

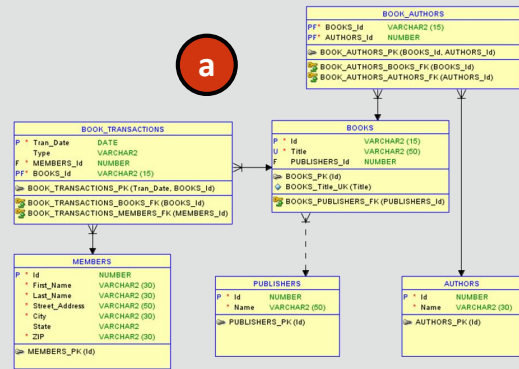
- d. Click the Relational tab to see the engineered Relational Model



Reverse Engineering a Relational Model

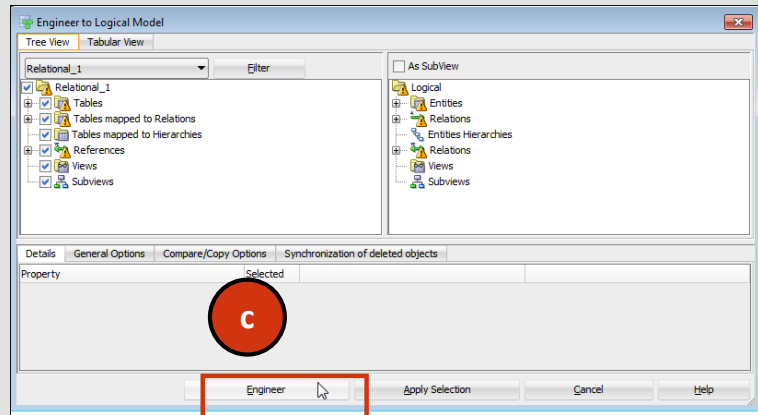
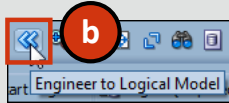
- Reverse engineering from a Relational to a Logical Model allows an ERD to be created from an existing Physical design
- Here are the steps involved in reverse engineering a relational model using Oracle SQL Developer Data Modeler:

a. Select the relational model



Reverse Engineering a Relational Model

- b. Click the Engineer to Logical Model icon
- c. Accept all the defaults and click Engineer



Project Exercise

- DFo_4_2_Project
 - Oracle Baseball League Store Database
 - Engineer the Relational Model from the Logical Data Model



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Summary

- In this lesson, you should have learned how to:
 - Describe how to convert a logical model to a relational model in Oracle SQL Developer Data Modeler
 - List the steps to convert a logical model to a relational model
 - List the steps to convert a relational model to a logical model in Oracle SQL Developer Data Modeler



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